DAY - 13 SEAT NUMBER | 0 | 0 | 0 | 0 | 6 | 5 |

2013	III	11	1100	V – 768	(E)

COMPUTER SCIENCE
PAPER - II (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : *(1) All questions are compulsory.*
(2) Figures to the right indicate full marks.
(3) Draw neat diagrams wherever necessary.
(4) Use of any type of calculator is not allowed.
(5) Due credit will be given for the programs with appropriate remarks or comments.

1. (A) Select correct alternatives and rewrite the following :

(a) ALU is _____ bit unit in 8085 Microprocessor. 1

(i) 8
(ii) 16
(iii) 32
(iv) 64

(b) The flag bit that gets affected on execution of RRC instruction in 8085 Processor is _____. 1

(i) Zero
(ii) Parity
(iii) Carry
(iv) All

(c) _____ is a Microcontroller Chip. 1

(i) 8085
(ii) 8086
(iii) 8051
(iv) Pentium

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(d) _____ cable type is ideal for connecting between two buildings. 1

(i) UTP

(ii) STP

(iii) Co-axial

(iv) Flat

(B) Answer **any two** of the following :

(a) Write a short note on Interrupts in 8085 Microprocessor. 3

(b) Explain any three addressing modes of 8085 Microprocessor with an example of each. 3

(c) Explain various applications of Microcontroller. 3

2. (A) Answer **any two** of the following :

(a) Explain functions of the following registers of 8085 Microprocessor : 3

(i) Accumulator

(ii) Flag

(iii) Program Counter

(b) Explain the following instructions of 8085 Microprocessor with suitable example of each : 3

(i) XTHL

(ii) DAD VP

(c) Explain the following characteristics of transmission media : 3

(i) Bandwidth

(ii) Band Usage

(iii) Attenuation

(B) Answer **any one** of the following :

(a) Draw block diagram of 8085 Microprocessor. 4

(b) Compare any four attributes of UTP and Optical Fiber Cable. 4

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3. (A) Answer **any two** of the following :

(a) The flag register of 8085 Microprocessor contains the data AAH. Interpret its meaning. 3

(b) Describe in brief the functions of the following pins of 8085 Microprocessor : 3

(i) IO/$\overline{M}$

(ii) SID

(iii) HLDA

(c) Define Bus, Ring and Star Topologies. Draw simple diagram for each. 3

(B) Answer **any one** of the following :

(a) Explain the following instructions with diagram : 4

(i) RLC

(ii) RRC

(iii) RAL

(iv) RAR

(b) Compare any four attributes of 80286 and 80486 Processors. 4

4. (A) Answer **any two** of the following :

(a) The accumulator of 8085 processor contains data B8H and register contains 44H. What will be the content of accumulator after execution of each of the following instructions independently ? 3

(i) ORI FOH

(ii) ANA B

(iii) XRI OFH

(b) Explain the following instructions of 8085 Microprocessor with suitable example of each : 3

(i) STD addr

(ii) ADD r

(iii) CMP r

(c) Explain Hub and Repeater in detail. 3

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(B) Answer **any one** of the following :

(a) Explain advantages of the following features of the Pentium Processor : 4

(i) Dual Pipelining

(ii) On Chip Caches

(iii) Branch Prediction

(iv) 64-bit Data Bus

(b) What is meant by Protocol ? Explain Concept of TCP/IP Protocol. 4

5. Answer **any two** of the following :

(a) Write an assembly language program to count number of even data bytes occurring in a block stored from memory location C051H and onwards. The length of block is stored in location C050H. Store result in location C060H. 5

(b) Write an assembly language program to perform multiplication of two 8-bit numbers where multiplicand is stored at the memory locations C051H and C052H and multiplier is stored at C053H. The result is to be stored at memory location address C054H to CO55H. 5

(**Note :** 8-bit multiplicand is extend to 16-bit)

(c) A flex number is stored at location 3000H. Write an assembly language program to interchange its digits. The new number is to be stored at 3001H. Add original number with new number and store result at location 3010H. 5

OR

5. Answer **any two** of the following :

(a) Write an assembly language program to count occurence of the data ABH in a memory block starting from 4000H to 400FH. Store count at memory location 4500H. 5

(b) A block of data is stored in memory locations from 7501H to 75FFH. Write an assembly language program to transfer block in reverse order to memory location 7600H and onwards. 5

(c) Write an assembly language program to find largest element in a block of data. The length is in memory location D001H and block begins in memory location D002H. Store maximum in D000H. Assume that all numbers are 8-bit unsigned binary numbers. 5